4.6.4 Data quality

Poor data quality in digital twins can lead to failures that compromise operational safety and business outcomes. Faulty sensor readings can trigger incorrect automated decisions, while corrupted data feeds may undermine predictive models and analytics. Establishing robust data quality standards requires defining acceptable ranges for each sensor type, validation rules for imported data sources, and standardized procedures for handling missing or anomalous values. These standards should account for both individual sensor constraints and cross-sensor relationships that can validate data plausibility.

4.7 Selecting the right combination

Making optimal decisions about data architecture requires understanding how data types, sources, structures, ingestion methods, and storage technologies work together. Table 4.2 illustrates common types, sources, and integration patterns for the data found in a digital twin.

Table 4.2 Typical sources of different data types in a digital twin and how these are typically structured, integrated, and stored.

Data type	Typical sources	Integration method	Structure	Storage technology	Considerations
Reference data	ERP, asset management	Batch, API	Relational	PostgreSQL, Memgraph	Update frequency, consistency
Timeseries data	IoT sensors, SCADA	Streaming	Columnar	InfluxDB, Timestream	Volume retention, query patterns
Spatial data	GIS, surveys	Batch, API	Specialized	PostGIS, Memgraph	Complexity, query types, accuracy requirements
Derived data	Analytics, ML	Batch processing	Various	Feature stores, S3	Refresh frequency, serving requirements
Unstructured data	Cameras, documents	API, batch	Binary/text	S3, blob storage	Size, access patterns, processing needs

This decision matrix highlights that there is no single 'best' data storage solution; rather, optimal architecture is data type-driven.

Transactional data utilizes relational databases for schema enforcement and consistency, which is crucial for ERP systems. In contrast, analytical data relies on specialized storage technologies optimized for rapid ingestion, high-volume querying, or complex indexing.